ParkViewRT: Vision-Based Parking Slot Monitoring System

Project ID: FYP02-CS-T2610-833

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DATA SCIENCE

Executive Summary

ParkViewRT is a vision-based parking slot monitoring system developed for the FCI and FAIE parking areas at Multimedia University. The system uses recorded parking videos, pretrained YOLOv11 vehicle detection, manually labelled parking slot polygons, and a web-based dashboard to display parking slot status.

The backend was developed using FastAPI, Python, OpenCV, Ultralytics YOLO, and Shapely. The frontend was developed using React, Vite, and TypeScript. The system supports FCI day, FCI night, FAIE day, and FAIE night video variants.

Objectives

1. Develop a local vision-based parking monitoring system for MMU FCI and FAIE.
2. Use parking slot polygon labels to map detected vehicles to specific slots.
3. Integrate a pretrained YOLO vehicle detector with a FastAPI backend.
4. Implement slot status logic using detection results, slot overlap, box overlap, and thresholds.
5. Develop a React dashboard showing occupied, available, occluded, and unmonitored slot states.
6. Evaluate the system using model comparison, threshold tuning, backend/frontend testing, API checks, debug validation, and verified slot labels.

Results & Discussions

Validation Dataset

- 4 video variants: FCI Day, FCI Night, FAIE Day, FAIE Night
- **11 evenly spaced frames per variant**
- Total validation frames: **44**
- Total verified monitored labels: **2,167**
- Label distribution:
  - **Available: 1,009**
  - **Occupied: 1,106**
  - **Occluded: 52**

Model Comparison

Model	Correct Labels	Mismatches	Accuracy
YOLO11n	2,167	**0**	**100.00%**
YOLO12n	2,140	**27**	**98.75%**
YOLO26n	2,070	**97**	**95.52%**
YOLOv8n	2,033	**134**	**93.82%**

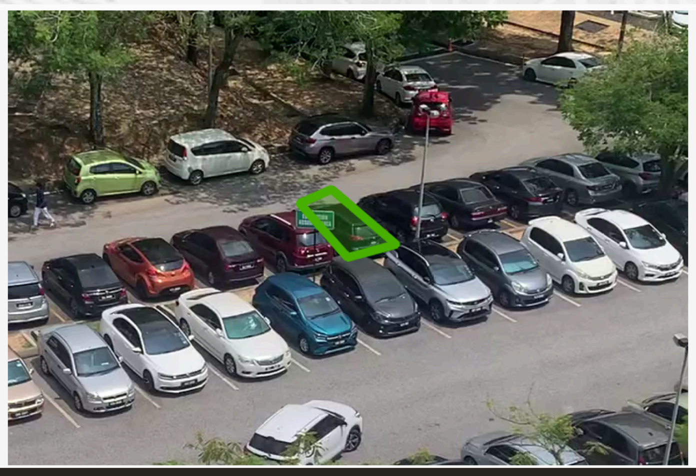
YOLO11n was the selected model because it classified all 2,167 verified labels correctly with 0 mismatches.

Threshold Tuning

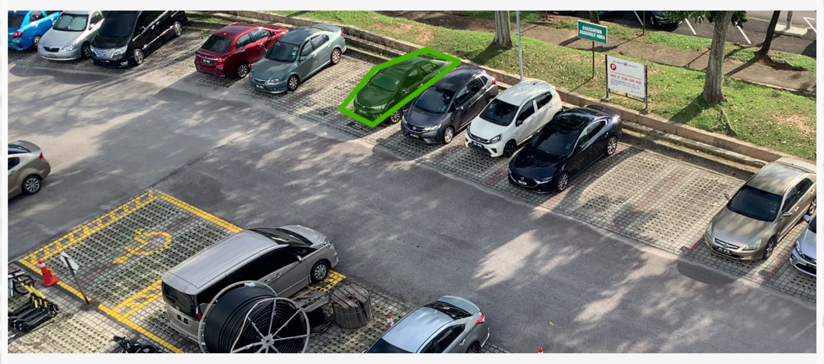
- Total combinations tested: **210**
- Selected confidence: **0.20**
- Selected slot overlap: **0.25**
- Selected box overlap: **0.20**
- **Final result: 100.00% accuracy**
- **False available: 0**
- **False occupied: 0**
- **False occluded: 0**

Limitations

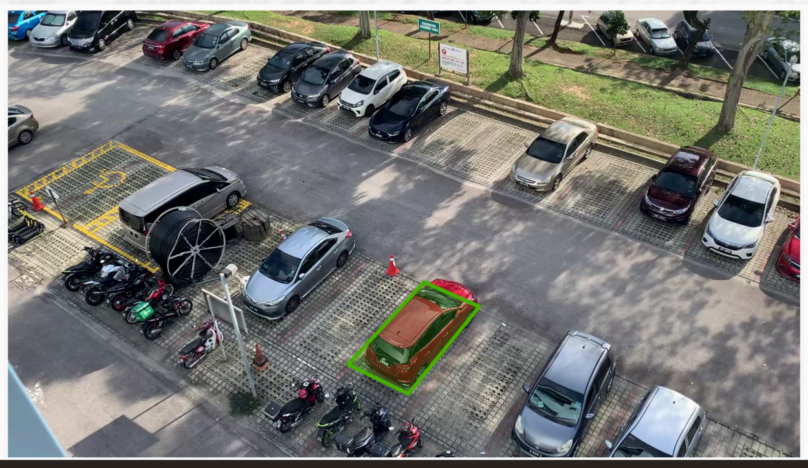
- The system works as a local prototype for selected MMU parking areas.
- Current input uses recorded videos, not live CCTV feeds.
- Slot accuracy depends on fixed camera angles and manually labelled polygons.
- Lighting, shadows, occlusion, and viewpoint limitations may affect wider deployment.



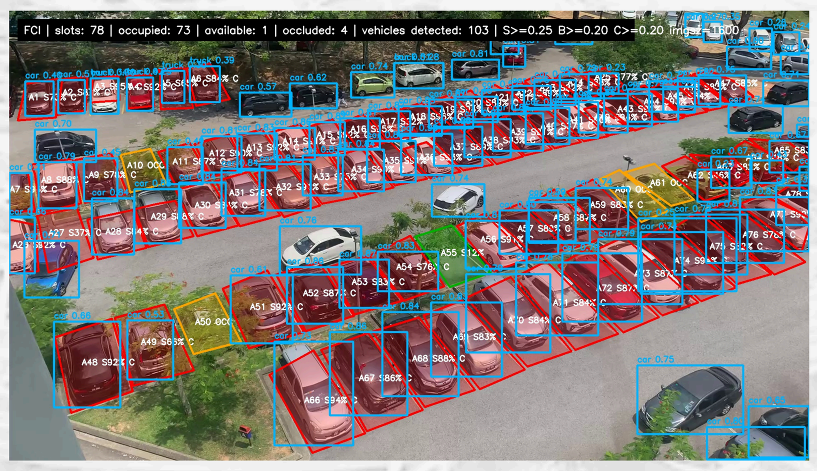
YOLO12n False Classification Example
FCI Day | Slot A20 | Frame 0
Expected: Occupied | Predicted: Available

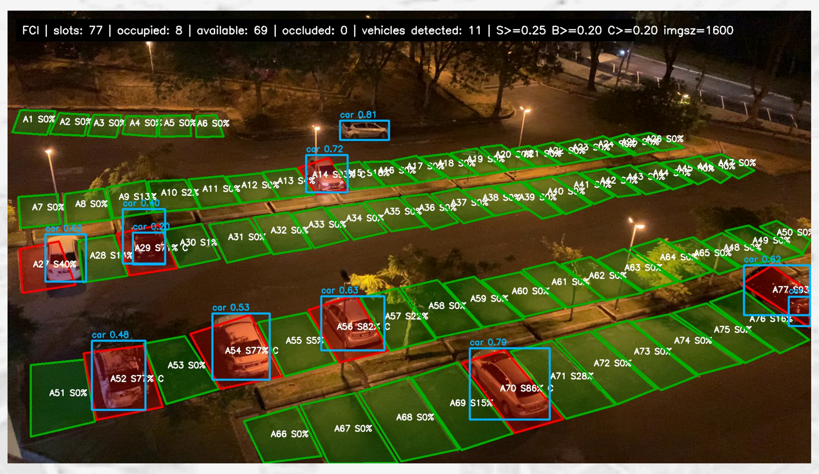


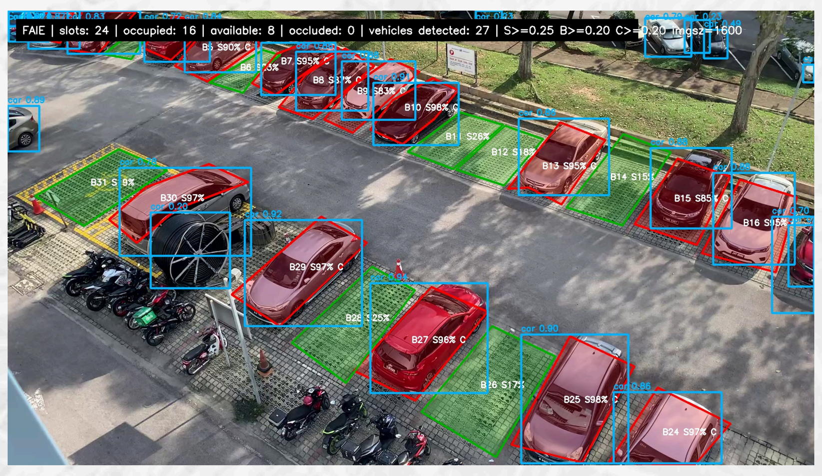
YOLO26n False Classification Example
FAIE Day | Slot B7 | Frame 7238
Expected: Occupied | Predicted: Available



YOLOv8n False Classification Example
FAIE Day | Slot B27 | Frame 0
Expected: Occupied | Predicted: Available







Achievements

1. FCI and FAIE parking dashboards were implemented using local recorded videos.
2. Runtime JSON slot polygon files were prepared for all the 4 variant videos
3. A pretrained YOLOv11 model was integrated with the FastAPI backend for vehicle detection.
4. Slot status was generated using polygon overlap, box overlap, centre-based checks, and predetermined thresholds.
5. The React dashboard displays video preview, status summary and colour-coded slot visualization.
6. Final evaluation used 44 validation frames and 2,167 manually verified monitored slot labels.

Future Work

1. Integrate live CCTV or IP camera feeds instead of relying only on recorded videos.
2. Expand the dataset with more parking videos, different weather conditions, lighting levels and camera angles.
3. Explore custom YOLO training or fine-tuning to improve detection for small, shaded, distant, or partially occluded vehicles.
4. Improve or automate parking slot polygon labelling to reduce manual setup effort.
5. Improve deployment so the system can be tested in more facility parking areas.

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FINAL YEAR PROJECT 2026

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